

THE STORY OF ANATOMY

THE SECRETS OF THE LIVING BODY HAVE LONG FASCINATED BOTH SCIENTISTS AND ARTISTS

The exploration of biological structure—**anatomy**—is the basis for understanding how bodies work. Early anatomists had to dissect cadavers to find answers to even simple questions; later, technologies such as the microscope helped physicians chart the body in greater detail.

In the ancient world, anatomists dissected the bodies of animals but were forbidden to open human cadavers, which were considered sacred. As a result, Galen (129–200 CE), Rome's most celebrated physician, circulated erroneous ideas

about the human body that were based on animal anatomy. When human dissection was sanctioned, Galen's ideas were corrected through direct

observation. By the Renaissance, artists such as Leonardo da Vinci (see 1468–82) were illustrating bodies with exquisite realism, and each new anatomical publication charted and named new structures. Flemish-born anatomist Andreas Vesalius (see 1543) dominated the scene with his illustrated *De Humani Corporis Fabrica*.

DELVING DEEPER

With the invention of microscopes in the 1600s, anatomists could see that organs were made up of cellular tissues. By the 1900s, the discovery of X-rays heralded new directions for anatomy. Today, powerful electron microscopes can probe the detailed structure of cells, and new imaging techniques reveal internal structures in 3-D without the need to cut the body open.



Anatomical waxwork

Three-dimensional figures, such as this 19th-century wax fetus, were important tools for teaching medicine.



PRESERVING ANATOMICAL SPECIMENS

Dead body parts decay quickly. Preservation in alcohol prolongs opportunities for study, but also dehydrates specimens, causing distortion. Formalin is commonly used as a fixative to avoid this. Some of the most sophisticated modern methods preserve bodies in a dry state, for example, by replacing water and fat with plastic.

“THE HUMAN FOOT IS A MASTERPIECE OF ENGINEERING AND A WORK OF ART.”

Leonardo da Vinci, Italian polymath, from his notebooks, 1508–18

